A direct proof of the complete Crouzeix inequality

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Abstract

Every square matrix A comes with a compact convex subset $W(A)$ of the complex plane, called its numerical range. The complete Crouzeix conjecture says: if you plug A into a matrix-valued polynomial P, the result is never larger than twice the biggest value of P on $W(A)$, and this is uniform in all matrix dimensions. We give a self-contained direct proof. It uses an exterior conformal map of an analytic convex domain containing $W(A)$, a positive kernel on the unit circle, and a two-by-two positive block comparison. Degenerate cases, where $W(A)$ is a single point or a line segment, are included.

1 Introduction

How to read this paper

The first section states the main theorem and explains what is going on. Sections 2–6 prove it. The Appendix collects three background items: the exterior conformal map, a short recall of Schwarz reflection, and the classical Jordan-domain theorem references.

If you are a strong high-school student, you already know enough linear algebra and complex numbers to read the Introduction and to see the main ideas in Section 5. The one technical part is the conformal map in Section 2 and Appendix A. You may treat Appendix A as a black box on a first read.

The numerical range

Fix a positive integer n and a matrix $A \in M_n(\mathbb{C})$. A vector $u \in \mathbb{C}^n$ with $u^*u = 1$ is a *unit vector*. Its *Rayleigh quotient* is the complex number u^*Au. The *numerical range* of A is the set

$$W(A) \;=\; \{\, u^*Au \;:\; u \in \mathbb{C}^n,\; u^*u = 1 \,\}. \tag{1.1}$$

The numerical range of A is always a compact convex subset of $\mathbb{C}$, and it always contains the spectrum of A: a unit eigenvector u of an eigenvalue λ satisfies $u^*Au = \lambda u^*u = \lambda$.

Two extreme examples are useful.

- If $A = \operatorname{diag}(\lambda_1, \dots, \lambda_n)$ is diagonal with real entries, $W(A)$ is the interval $[\min \lambda_i,\ \max \lambda_i]$ on the real axis.
- If $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, then $|u^*Au| = 2|u_1u_2| \le |u_1|^2 + |u_2|^2 = 1$ for every unit u, with equality whenever $|u_1| = |u_2| = 1/\sqrt{2}$. Convexity then forces $W(A)$ to be the closed unit disk $\overline{\mathbb{D}}$. Notice that the spectrum is just $\{0\}$, yet the numerical range is the whole disk. This is the simplest example where nonnormality shows up in $W(A)$ but not in the spectrum.

For a normal matrix (one with $AA^* = A^*A$), the spectral theorem gives a unitary U with $A = U\Lambda U^*$, and then u^*Au is a convex combination of the eigenvalues. So $W(A)$ is exactly the convex hull of the eigenvalues. For a general matrix, this statement fails badly: the numerical range is genuinely larger than the spectrum, and that extra size is what makes the spectral-set constant 2 possible.

The main theorem

We allow *matrix-valued* polynomials. If $P(z) = \sum_{k=0}^{d} B_k z^k$ with coefficient matrices $B_k \in M_m(\mathbb{C})$, define

$$P[A] \;=\; \sum_{k=0}^{d} A^k \otimes B_k \;\in\; M_n(\mathbb{C}) \otimes M_m(\mathbb{C}) \;\cong\; M_{nm}(\mathbb{C}). \tag{1.2}$$

Here $\otimes$ is the Kronecker product: the factor A^k acts on $\mathbb{C}^n$ and the factor B_k acts on $\mathbb{C}^m$, and the two actions commute. Equivalently, $P[A]$ is obtained by replacing the scalar variable z with the matrix A, keeping the coefficients B_k in the exact order written. The two tensor factors are the *base* $\mathbb{C}^n$ and the *coefficient* $\mathbb{C}^m$.

All norms are the operator norms from the Euclidean inner products and their tensor products (equivalently, the largest singular value). The Hilbert–Schmidt norm of a matrix C is denoted $\|C\|_2 = (\operatorname{tr} C^*C)^{1/2}$.

Theorem 1.1 (Complete Crouzeix inequality)**.** *Let $n, m \geq 1$, let $A \in M_n(\mathbb{C})$, let $d \geq 0$, and let $P(z) = \sum_{k=0}^{d} B_k z^k$ with $B_k \in M_m(\mathbb{C})$. Then*

$$\|P[A]\| \;\leq\; 2 \max_{z \in W(A)} \|P(z)\|. \tag{1.3}$$

The constant 2 is optimal: it cannot be lowered uniformly in n, m, d.

The word *complete* refers to the uniformity in the coefficient dimension m. A scalar proof (the case $m = 1$) does not automatically extend, because scalar coefficients commute but matrix coefficients need not: $F(z)G(z)$ and $G(z)F(z)$ are different $M_m(\mathbb{C})$-valued functions in general, and $F[A]G[A]$ is not $G[A]F[A]$. The proof below keeps these two products separate all the way through, which is why it works uniformly in m.

Why the constant cannot be less than 2

Take $n = 2$, $m = 1$, $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$, and $P(z) = z$. The eigenvalue 0 of A has a two-dimensional generalised eigenvector space, and $\|A\| = 2$. Meanwhile the previous subsection showed that $W(A)$ is the closed unit disk, so $\max_{W(A)} |P(z)| = \max_{W(A)} |z| = 1$. The ratio $\|P[A]\| / \max_{W(A)} |P| = 2$, so no universal constant below 2 works. The rest of the paper proves that 2 is enough.

Roadmap of the proof

The argument has five steps.

1. **Enclose.** Place $W(A)$ strictly inside a bounded convex open set Ω whose boundary is a smooth real-analytic Jordan curve. This lets us use conformal mapping. The final section shows how to remove this hypothesis.

2. **Coordinate.** Build an exterior conformal map $h : \{\lambda : |\lambda| > 1\} \to \mathbb{C} \setminus \overline{\Omega}$. On the unit circle $|\lambda| = 1$, h parametrises $\partial\Omega$, so every boundary integral becomes a circle integral.

3. **Kernel.** Expand the Cauchy kernel $q(\lambda)/(h(\lambda) - z)$ at infinity, where $q(\lambda) = \lambda h'(\lambda)$. This produces polynomials $b_k(z)$ (the *Faber polynomials* of the exterior coordinate), and a positive kernel $K(\lambda, \mu)$ on the unit circle. Positivity of K is where convexity of Ω enters the proof.

4. **Coefficient bound.** From K, deduce that for every matrix-valued polynomial $G = \sum b_k C_k$,

$$\sum_k \|C_k\|_2^2 \;\le\; \int_{\mathbb{T}} \|G(h(\mu))\|_2^2 \, d\tau(\mu) \;\le\; \|C_0\|_2^2 + 2 \sum_{k \ge 1} \|C_k\|_2^2. \tag{1.4}$$

   The factor of 2 on the right is the sole source of the constant 2 in Theorem 1.1. This step uses the positivity of K in a decisive way.

5. **Functional analysis.** Let $R(\lambda) = q(\lambda)(h(\lambda)I - A)^{-1}$ be the exterior resolvent. Positivity of $R + R^*$ (a consequence of $W(A) \subset \Omega$) lets a two-by-two positive block comparison pass (1.4) to the single polynomial $F[A]$ and force $\|F[A]\| \le 2$ whenever $\|F\| \le 1$ on $\partial\Omega$. Then shrink Ω down to $W(A)$.

Prerequisites and notation

We use:

- Linear algebra: unitary matrices, tensor products, singular values, operator norm $\|\cdot\|$ and Hilbert–Schmidt norm $\|\cdot\|_2$.
- Complex analysis: holomorphic functions, conformal maps, the Riemann mapping theorem, Schwarz reflection. A short self-contained proof of the exterior collar is in Appendix A.
- Fourier analysis on the unit circle $\mathbb{T} = \{|z| = 1\}$, equipped with the normalised measure $d\tau(e^{it}) = dt/(2\pi)$. Every constant function of modulus one has integral one; every function with only positive or only negative Fourier modes has integral zero.

Historical context

Crouzeix asked the scalar question in 2004 and proved a universal complete bound of 11.08 in 2007 [7, 8]. The exterior-kernel method traces back to Delyon and Delyon [10]. Crouzeix and Palencia obtained the complete bound $1 + \sqrt{2}$ [9], and Ransford and Schwenninger simplified the final step [16]. Badea, Crouzeix and Delyon proved the complete constant 2 for all base matrices of order 2 [3], and Åhag, Czyż, Virtanen proved the complete constant 2 for base matrices of order 3 [1]. The scalar case $m = 1$ was settled recently by Jin [12], Lorist and Schwenninger [13], and Luo [14]. A structural proof of the complete case is given in the companion preprint [15]; the exterior kernel, marginal identities and positive block comparison below have counterparts there. The point of the present paper is to give a direct and self-contained argument for the *complete* case, with all matrix dimensions independent.

2 Exterior coordinates and Faber coefficients

Throughout this section, $\Omega \subset \mathbb{C}$ is a bounded open convex set whose boundary is a *regular real-analytic Jordan curve*: a simple closed curve parametrised real-analytically with nowhere-zero derivative. The unit circle is $\mathbb{T} = \{\lambda : |\lambda| = 1\}$. The exterior unit disk is $\mathbb{C} \setminus \overline{\mathbb{D}} = \{\lambda : |\lambda| > 1\}$. Every circle integral uses $d\tau(e^{it}) = dt/(2\pi)$, so constant functions of modulus one have integral one.

The exterior coordinate

By Lemma A.1 in the Appendix, there is a number $0 < r_0 < 1$ and a map

$$h(\lambda) \;=\; c\lambda + c_0 + \sum_{j\geq 1} c_j \lambda^{-j}, \qquad c \neq 0, \tag{2.1}$$

meromorphic on $\{\lambda : |\lambda| > r_0\} \cup \{\infty\}$, one-to-one there as a map of the Riemann sphere, with a simple pole at infinity, mapping $|\lambda| > 1$ conformally onto $\mathbb{C} \setminus \overline{\Omega}$ and $|\lambda| = 1$ onto $\partial\Omega$. This is the *exterior conformal map*, or *exterior coordinate* of Ω.

The Appendix gives a complete derivation from Riemann's mapping theorem. The reader who wants to accept it can take (2.1) and the statements below as basic inputs.

The supporting line and the outward normal

Define

$$q(\lambda) \;=\; \lambda h'(\lambda). \tag{2.2}$$

The two partial derivatives of h at a point $e^{i\theta} \in \mathbb{T}$ are $\partial_\theta h = iq$ (tangent to the boundary) and $\partial_r h = q$ where $r = |\lambda|$ (radial direction, pointing outward from $\overline{\Omega}$). So $q(\lambda)$ is a nonzero outward normal vector at $h(\lambda) \in \partial\Omega$, and $iq(\lambda)$ is a positively oriented tangent vector.

Because Ω is convex, the entire set $\overline{\Omega}$ lies on the inner side of the supporting line through $h(\lambda)$ with normal direction $q(\lambda)$:

$$\Re\Big(\overline{q(\lambda)}\big(h(\lambda) - z\big)\Big) \;\geq\; 0 \qquad (\lambda \in \mathbb{T},\ z \in \overline{\Omega}). \tag{2.3}$$

This is the only geometric input from convexity, and it is what makes the kernel below nonnegative. Figure 1 shows the picture.

$$\begin{array}{ccc} \mathbb{T} \text{ exterior} & \xrightarrow{\;h\;} & \text{exterior of } \overline{\Omega} \\ \lambda & & h(\lambda),\ \text{outward normal } q(\lambda) \end{array}$$

Figure 1: Schematic. The exterior conformal map h sends the exterior unit circle onto the exterior of $\overline{\Omega}$, one-to-one and onto. At $\lambda \in \mathbb{T}$, the vector $q(\lambda) = \lambda h'(\lambda)$ points outward from $\overline{\Omega}$. Convexity of Ω forces every $z \in \overline{\Omega}$ to lie on the inner side of the supporting line $\Re(\overline{q(\lambda)}(w - h(\lambda))) = 0$.

Faber polynomials

Define scalar polynomials $b_k(z)$, $k \geq 0$, by the expansion at infinity

$$\frac{q(\lambda)}{h(\lambda) - z} \;=\; \sum_{k\geq 0} b_k(z)\, \lambda^{-k}, \qquad |\lambda| \gg 1. \tag{2.4}$$

The left side is holomorphic at $\lambda = \infty$ (since $h(\lambda) \sim c\lambda$ with $c \neq 0$), so the expansion makes sense. These b_k are the *Faber polynomials* of the exterior coordinate h; they are the natural generalisation of the monomials z^k to the domain Ω. In the simplest case $\Omega = \mathbb{D}$ with $h(\lambda) = \lambda$, one has $b_k(z) = z^k$; for a general Ω, the b_k have corrections coming from the shape of $\partial\Omega$.

We can be more precise: after factoring $c\lambda$ from the denominator, we find $b_0 = 1$ and, for $k \geq 1$, b_k is a polynomial of degree k with leading coefficient c^{-k}. So $(b_k)_{k\geq 0}$ really is a basis of the space of polynomials, and polynomial degrees match indices. This is useful because it lets us rewrite any matrix polynomial $G(z) = \sum_{k=0}^{N} b_k(z)C_k$ uniquely in the Faber basis, with $C_k \in M_m(\mathbb{C})$ in the natural way.

Boundary values and the mixed coefficient

The boundary restrictions $b_k \circ h$ of the Faber polynomials have a remarkably clean Fourier form. There are scalars s_{kj}, indexed by $k, j \geq 1$, such that on the exterior collar one has

$$b_k(h(\mu)) \;=\; \mu^k \;+\; \sum_{j\geq 1} s_{kj}\, \mu^{-j}, \qquad k \geq 1. \tag{2.5}$$

Equivalently, the *mixed correction kernel*

$$s(\lambda, \mu) \;=\; \frac{q(\lambda)}{h(\lambda) - h(\mu)} \;-\; \frac{\lambda}{\lambda - \mu} \;=\; \sum_{k,j\geq 1} s_{kj}\, \lambda^{-k}\mu^{-j} \tag{2.6}$$

collects exactly the difference between the true boundary kernel and the boundary kernel of the unit disk.

The function s has a nice structure. It extends holomorphically across the diagonal $\lambda = \mu$ and across both infinities, so it is regular on a full product collar of the exterior. Its mixed Taylor coefficients decay geometrically:

$$|s_{kj}| \;\leq\; C_\rho\, \rho^{k+j} \qquad (r_0 < \rho < 1,\; k, j \geq 1). \tag{2.7}$$

Consequently the double series converges normally on the exterior product collar, hence absolutely and uniformly on $\mathbb{T}^2$. So every interchange of sum and integral below is legal.

The kernel we will need is

$$K(\lambda, \mu) \;=\; 1 + s(\lambda, \mu) + \overline{s(\lambda, \mu)}. \tag{2.8}$$

Since s has only strictly mixed Fourier modes ($\lambda^{-k}\mu^{-j}$ with $k, j \geq 1$), neither s nor its conjugate contributes to the integral in λ alone or in μ alone. So

$$\int_{\mathbb{T}} K(\lambda, \mu)\, d\tau(\lambda) = 1 \quad \text{(for each fixed } \mu\text{)}, \qquad \int_{\mathbb{T}} K(\lambda, \mu)\, d\tau(\mu) = 1 \quad \text{(for each fixed } \lambda\text{)}.$$

These are the *marginal identities*. They say: K is like the kernel of the identity operator, but with a real, symmetric, nonnegative correction.

The positive kernel theorem

Theorem 2.1 (Exterior Faber coefficients and the positive kernel)**.** *Let Ω be a bounded convex domain with regular real-analytic Jordan boundary. Let h be as in (2.1), and let q, $(b_k)_{k\geq 0}$, and s_{kj} be defined by (2.2), (2.4), and (2.6). Then:*

1. *$b_0 = 1$, and for each $k \geq 1$, b_k is a polynomial of degree k with leading coefficient c^{-k} in (2.1).*

2. *The mixed coefficients s_{kj} satisfy the geometric decay estimate (2.7).*

3. *The kernel K in (2.8) is continuous on $\mathbb{T}^2$, nonnegative everywhere, and has integral one in each variable with the other fixed.*

Proof. Part 1 (degree and leading coefficient). Write $h(\lambda) - z = c\lambda \cdot (1 + (c_0 - z)/(c\lambda) + O(\lambda^{-2}))$. For large $|\lambda|$,

$$\frac{q(\lambda)}{h(\lambda) - z} = \frac{c\lambda + O(\lambda^{-1})}{c\lambda} \cdot \left(1 + \frac{c_0 - z}{c\lambda} + \cdots\right)^{-1} = 1 + O(\lambda^{-1}).$$

Expanding the inverse as a geometric series in $(c_0 - z)/(c\lambda)$ produces, for the coefficient of λ^{-k}, a polynomial in z of degree k. Inspecting the largest power of z shows it is $c^{-k}z^k$. So $b_0 = 1$ and $b_k(z) = c^{-k}z^k + (\text{lower degree})$ for $k \geq 1$. In particular the b_k form a basis of the scalar polynomials.

Part 2 (geometric decay and joint extension). Consider

$$H(\lambda, \mu) \;=\; \frac{h(\lambda) - h(\mu)}{\lambda - \mu}. \tag{2.9}$$

Write $u = \lambda^{-1}$, $v = \mu^{-1}$, and $h(\lambda) = c\lambda + g(u)$ with g holomorphic on $|u| < r_0^{-1}$. Then

$$H(\lambda, \mu) = c - uv\,\frac{g(u) - g(v)}{u - v}.$$

The divided difference $(g(u) - g(v))/(u - v)$ is jointly holomorphic in (u, v), with diagonal value $g'(u)$, so this formula covers the diagonal and infinities. Direct differentiation gives

$$s(\lambda, \mu) \;=\; \frac{\lambda \partial_\lambda H}{H} \;=\; -\frac{u \partial_u H}{H}.$$

Since H is nonzero on the reciprocal bidisk $|u|, |v| < r_0^{-1}$ (injectivity of h, and $H \to c$ at infinity), s is holomorphic there and vanishes on both axes $\{u = 0\}$ and $\{v = 0\}$. Its Taylor expansion therefore contains exactly the strictly mixed powers $\lambda^{-k}\mu^{-j}$ with $k, j \geq 1$.

For any $r_0 < \rho < 1$, apply Cauchy estimates on the closed disk $|u|, |v| \leq \rho^{-1}$: there is a constant C_ρ with $|s_{kj}| \leq C_\rho \rho^{k+j}$. Summing the resulting geometric series shows the Laurent expansion (2.6) converges normally (absolutely and uniformly on compacta, including those meeting infinity) throughout the product collar $|\lambda|, |\mu| > r_0$. Restricting to $|\lambda| = |\mu| = 1$ gives absolute uniform convergence on $\mathbb{T}^2$.

Comparing (2.4) and (2.6) coefficient-by-coefficient in λ^{-k}, with μ fixed in the exterior collar, yields (2.5). Indeed, $\lambda/(\lambda - \mu) = \sum_{k \geq 0} \mu^k \lambda^{-k}$, so the difference of the two sides matches $b_k(h(\mu)) - \mu^k$.

Part 3 (positivity of K and marginal identities). For distinct $\lambda, \mu \in \mathbb{T}$, split K as

$$K(\lambda, \mu) \;=\; 2\,\Re \frac{q(\lambda)}{h(\lambda) - h(\mu)} \;+\; \left(1 - \frac{\lambda}{\lambda - \mu} - \frac{\bar{\lambda}}{\bar{\lambda} - \bar{\mu}}\right).$$

On $|\lambda| = |\mu| = 1$, we have $\bar{\lambda} = 1/\lambda$, $\bar{\mu} = 1/\mu$, and one verifies directly that $\lambda/(\lambda-\mu)+\bar{\lambda}/(\bar{\lambda}-\bar{\mu}) = 1$, so the bracketed term vanishes. Hence

$$K(\lambda, \mu) \;=\; 2\,\Re\frac{q(\lambda)}{h(\lambda) - h(\mu)}.$$

Now $\Re(q/(h(\lambda) - h(\mu))) = |q|^2/\Re(\overline{q}(h(\lambda) - h(\mu)))$ where $|q|^2 > 0$, and the denominator is nonnegative by (2.3) with $z = h(\mu)$. So $K(\lambda, \mu) \geq 0$ for distinct λ, μ, and continuity gives the same on the diagonal.

For the marginal identities: every term in s has Fourier mode $\lambda^{-k}\mu^{-j}$ with $k, j \geq 1$, so integrating against $d\tau(\lambda)$ picks out $k = 0$, which never occurs; and similarly for the conjugate. Hence $\int_{\mathbb{T}} s(\lambda, \mu)d\tau(\lambda) = \int_{\mathbb{T}} \overline{s(\lambda, \mu)}d\tau(\lambda) = 0$, so $\int_{\mathbb{T}} K(\lambda, \mu)d\tau(\lambda) = 1$. Symmetric argument in μ. □

Identification of b_k as the Faber polynomial

Let $\Phi = h^{-1}$ denote the inverse of h on the exterior collar $\mathbb{C} \setminus \overline{\Omega} \to \{\lambda : |\lambda| > 1\}$. Then (2.5), which continues to hold throughout the exterior collar, gives

$$b_k(z) - \Phi(z)^k = O(z^{-1}), \qquad z \to \infty.$$

So b_k is the polynomial part of Φ^k – the classical *Faber polynomial* associated with the exterior coordinate h (see [4, 5] for background). This is why the notation b_k is standard.

The mixed coefficients s_{kj} are closely related to the differentiated *Grunsky coefficients* of the exterior map Φ; the classical normalization gives a weighted coefficient inequality [6, eq. (1.1)–(1.3)]. The positivity estimate we need below is a consequence of convexity and the marginal identities already proved in Theorem 2.1; no extra Grunsky estimate is required.

3 The coefficient norm estimate

The positive kernel K from Theorem 2.1 is the entire source of the constant 2 in the main theorem. This section extracts one clean inequality from it. Fix any dimension m and any finite-family of coefficient matrices $C_0, C_1, \ldots, C_N \in M_m(\mathbb{C})$, and put

$$G(z) \;=\; \sum_{k=0}^{N} b_k(z)\, C_k.$$

That is, G is an $M_m(\mathbb{C})$-valued polynomial written in the Faber basis of Ω. We want to bound the Hilbert–Schmidt L^2-norm of $G \circ h$ on the unit circle from above and below by the coefficient Hilbert–Schmidt norms $\|C_k\|_2$.

The operator T defined by K

Consider the bounded operator

$$(Tu)(\mu) \;=\; \int_{\mathbb{T}} K(\lambda, \mu)\, u(\lambda)\, d\tau(\lambda), \qquad u \in L^2(\mathbb{T}; \mathcal{H}), \tag{3.1}$$

where $\mathcal{H}$ is any separable Hilbert space (in practice $\mathcal{H} = M_m(\mathbb{C})$ with the Hilbert–Schmidt inner product).

Lemma 3.1. *The operator T extends to a contraction of $L^2(\mathbb{T};\mathcal{H})$.*

Proof. Nonnegativity $K \geq 0$ and the marginal identity $\int_{\mathbb{T}} K(\lambda,\mu)\,d\tau(\lambda) = 1$ together give, for each fixed μ, by Cauchy–Schwarz,

$$\|(Tu)(\mu)\|^2 \;\leq\; \left(\int_{\mathbb{T}} K(\lambda,\mu)\,d\tau(\lambda)\right)\left(\int_{\mathbb{T}} K(\lambda,\mu)\,\|u(\lambda)\|^2\,d\tau(\lambda)\right) \;=\; \int_{\mathbb{T}} K(\lambda,\mu)\,\|u(\lambda)\|^2\,d\tau(\lambda).$$

Integrate over μ and use the other marginal identity:

$$\|Tu\|_{L^2}^2 \;\leq\; \int_{\mathbb{T}}\left(\int_{\mathbb{T}} K(\lambda,\mu)\,d\tau(\mu)\right)\|u(\lambda)\|^2\,d\tau(\lambda) \;=\; \|u\|_{L^2}^2.$$

□

What T does on monomials

For a vector-valued function of the form $u(\lambda) = \sum_{k=1}^{N} C_k\lambda^k$, where each $C_k \in M_m(\mathbb{C})$, direct substitution in (3.1) shows that only the s summand of K contributes: the term 1 in K gives a zero integral (nonzero frequencies $\lambda^1,\dots,\lambda^N$), and the term $\overline{s}$ gives a zero integral because $\overline{s(\lambda,\mu)} = \sum_{k,j\geq 1}\overline{s_{kj}}\lambda^k\mu^j$, whose product with $C_k\lambda^k$ has only nonnegative powers of λ. So

$$(Tu)(\mu) \;=\; \sum_{j\geq 1}\Big(\sum_{k=1}^{N} s_{kj}\,C_k\Big)\,\mu^{-j}. \tag{3.2}$$

The normal convergence in Theorem 2.1 makes all interchanges legal. Because the μ^{-j} are orthonormal in $L^2(\mathbb{T})$,

$$\|Tu\|_{L^2}^2 \;=\; \sum_{j\geq 1}\Big\|\sum_{k=1}^{N} s_{kj}\,C_k\Big\|_2^2,$$

where $\|\cdot\|_2$ denotes the Hilbert–Schmidt norm on $M_m(\mathbb{C})$. Combined with Lemma 3.1,

$$\sum_{j\geq 1}\Big\|\sum_{k=1}^{N} s_{kj}\,C_k\Big\|_2^2 \;\leq\; \sum_{k=1}^{N}\|C_k\|_2^2. \tag{3.3}$$

The two-sided coefficient inequality

Now observe that on the boundary circle, by (2.5),

$$G(h(\mu)) \;=\; \sum_{k=0}^{N} b_k(h(\mu))C_k \;=\; C_0 + \sum_{k=1}^{N} C_k\mu^k + \sum_{k=1}^{N}\left(\sum_{j\geq 1} s_{kj}\,\mu^{-j}\right)C_k \;=\; C_0 + u(\mu) + (Tu)(\mu).$$

The three summands C_0, u, and Tu have disjoint Fourier supports (constant, positive, negative). Parseval's identity applies:

$$\int_{\mathbb{T}}\|G(h(\mu))\|_2^2\,d\tau(\mu) \;=\; \|C_0\|_2^2 + \sum_{k=1}^{N}\|C_k\|_2^2 + \|Tu\|_{L^2}^2.$$

Combine this identity with (3.3). The case gives

$$\|C_0\|_2^2 + \sum_{k=1}^{N}\|C_k\|_2^2 \;\leq\; \int_{\mathbb{T}}\|G(h(\mu))\|_2^2\,d\tau(\mu) \;\leq\; \|C_0\|_2^2 + 2\sum_{k=1}^{N}\|C_k\|_2^2.$$

This is the key estimate. In words: passing from a coefficient array $(C_0, C_1, \dots)$ to the boundary values of G can enlarge the coefficient Hilbert–Schmidt norm by at most a universal factor of 2, and never shrinks it.

Proposition 3.2 (The coefficient norm estimate)**.** *For every positive integer m, every integer $N \geq 0$, and every $M_m(\mathbb{C})$-valued polynomial $G(z) = \sum_{k=0}^{N} b_k(z)\, C_k$ in the Faber basis of Ω, one has*

$$\sum_{k=0}^{N} \|C_k\|_2^2 \;\leq\; \int_{\mathbb{T}} \|G(h(\mu))\|_2^2 \, d\tau(\mu) \;\leq\; \|C_0\|_2^2 + 2 \sum_{k=1}^{N} \|C_k\|_2^2. \tag{3.4}$$

Proof. This is exactly the combined identity above. The lower bound uses only $\|Tu\|^2 \geq 0$. The upper bound uses Lemma 3.1 to shoehorn $\|Tu\|_{L^2} \leq \sum_k \|C_k\|_2$ into the Parseval identity. □

A note on the literature

For comparison with [15, Proposition 3.1], the companion paper's exterior map G and variables t, w correspond to h, λ, μ, respectively. The mixed correction satisfies $s(\lambda, \mu) = B(\mu, \lambda)$, so $s_{kj} = \beta_{jk}$ in that paper's notation. Our formula (3.2) uses this orientation, and the proof above does not require symmetry of the kernel.

4 The resolvent and the double-layer positivity

From now on the matrix $A \in M_n(\mathbb{C})$ is fixed, with $W(A) \subset \Omega$. We will also fix the exterior coordinate h of Ω from section 2. The purpose of this section is to define the *exterior resolvent* and show that it has a positive Hermitian part.

The resolvent function

Define, on the exterior circle $\mathbb{T}$,

$$R(\lambda) \;=\; q(\lambda)\,(h(\lambda)I - A)^{-1} \;\in\; M_n(\mathbb{C}). \tag{4.1}$$

Here $h(\lambda)I - A$ is invertible whenever $h(\lambda) \notin \sigma(A)$. Since $\sigma(A) \subset W(A) \subset \Omega$ (compact) and the circle $\mathbb{T}$ parametrises $\partial\Omega$ (compact, disjoint from $\sigma(A)$), the resolvent exists and is continuous on $\mathbb{T}$, and extends holomorphically to a collar $\{|\lambda| > r_A\}$ for some $r_0 < r_A < 1$ with r_A closer to one than r_0.

Theorem 4.1 (Resolvent expansion and double-layer positivity)**.** *Let $\Omega, h, q, (b_k)$ be as above. For every $n \geq 1$ and every $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$, the following hold.*

1. Expansion. *R has the Laurent series*

$$R(\lambda) \;=\; \sum_{k \geq 0} b_k(A)\, \lambda^{-k}, \qquad |\lambda| \geq 1, \tag{4.2}$$

converging absolutely and uniformly on $\mathbb{T}$. The coefficient matrices $b_k(A)$ decay geometrically: there are constants $C < \infty$ and $0 < \theta < 1$ (depending on A and h) with $\|b_k(A)\| \leq C\theta^k$ for all $k \geq 0$.

2. Means. *$\int_{\mathbb{T}} R(\lambda)\, d\tau(\lambda) = I$ and $\int_{\mathbb{T}} (R(\lambda) + R(\lambda)^*)\, d\tau(\lambda) = 2I$.*

3. Positivity. *On the exterior circle,* $R(\lambda) + R(\lambda)^* \succeq 0$.

Proof. Part 1 (expansion and decay). Every eigenvalue of A belongs to $W(A) \subset \Omega$, by testing the corresponding unit eigenvector. Hence the resolvent $(h(\lambda)I - A)^{-1}$ is holomorphic on and outside $\mathbb{T}$, including a small exterior collar. Expanding at ∞ in λ gives a Laurent series in λ^{-1}, and comparing with (2.4) with z replaced by A shows that the coefficient of λ^{-k} is exactly $b_k(A)$. The series converges absolutely and uniformly on $\mathbb{T}$ by compactness of the exterior collar. To get the geometric bound, pick $r_A < \theta < 1$ with R holomorphic on $|\lambda| > r_A$, apply Cauchy estimates on the circle $|\lambda| = \theta^{-1}$.

Part 2 (means). The constant coefficient $b_0(A) = b_0(z)\big|_{z=A} = I$ (Theorem 2.1, part 1). So by (4.2) and orthogonality of the modes λ^{-k}, $\int_{\mathbb{T}} R\, d\tau = b_0(A) = I$. Taking adjoints and adding gives $\int_{\mathbb{T}}(R + R^*)\, d\tau = 2I$.

Part 3 (positivity). Let $B(\lambda) = h(\lambda)I - A$ so that $R = qB^{-1}$. Then $R^* = \overline{q}\,(B^*)^{-1}$ and

$$R + R^* \;=\; q\,B^{-1} + \overline{q}\,(B^*)^{-1} \;=\; (B^*)^{-1}\,(\overline{q}\,B + q\,B^*)\,B^{-1}.$$

Since $(B^*)^{-1}$ and B^{-1} are adjoints of each other, they give a congruence that preserves positivity. So it suffices to show that $\overline{q(\lambda)}\,B(\lambda) + q(\lambda)\,B(\lambda)^* \succeq 0$.

For any unit vector $v \in \mathbb{C}^n$, the quadratic form on the left at v is

$$v^*\,(\overline{q}\,B + q\,B^*)\,v \;=\; \overline{q}\,v^*Bv + q\,v^*B^*v \;=\; 2\Re\Big(\overline{q(\lambda)}\,v^*B(\lambda)v\Big)\,.$$

Substituting $B = h(\lambda)I - A$:

$$v^*Bv \;=\; h(\lambda) - v^*Av.$$

So the quadratic form is $\Re\Big(\overline{q(\lambda)}(h(\lambda) - v^*Av)\Big)$. By the support inequality (2.3) with $z = v^*Av \in W(A) \subset \overline{\Omega}$, this is ≥ 0. The congruence argument finishes the proof. □

5 The main estimate on an analytic convex domain

Now the payoff. In this section $\Omega, h, q, (b_k)$ are as in Section 2, $A \in M_n(\mathbb{C})$ satisfies $W(A) \subset \Omega$, and $R(\lambda) = q(\lambda)(h(\lambda)I - A)^{-1}$ is the exterior resolvent from Theorem 4.1.

Statement and strategy

Theorem 5.1 (Bound on an analytic convex domain)**.** *Let $n, m \geq 1$. Let $\Omega \subset \mathbb{C}$ be bounded, open, convex, with regular real-analytic Jordan boundary, and let $A \in M_n(\mathbb{C})$ with $W(A) \subset \Omega$. Let F be an $M_m(\mathbb{C})$-valued polynomial with $\|F(z)\| \leq 1$ on $\partial\Omega$ (equivalently, on $\overline{\Omega}$, since F is a polynomial and the boundary controls the closure). Then*

$$\|F[A]\| \;\leq\; 2.$$

The proof is a bookkeeping exercise with three moving pieces:

- Top singular vectors x, y of $F[A]$ that turn the norm into a *scalar* $y^*F[A]x = \|F[A]\|$.
- The two products FG and GF that turn $x^*G[A]x$ and $y^*G[A]y$ into the same scalar $y^*(\cdot)[A]x$ for two different choices of G.
- The two-by-two block $R + R^* \succeq 0$ that lets positivity compare the three fields $P + P^*$, $Q + Q^*$, and $M + M^*$ to each other.

Each piece contributes one inequality, and together they force $\|F[A]\| \leq 2$.

Setup: singular vectors and block notation

Put $\gamma = \|F[A]\|$. If $\gamma = 0$, there is nothing to prove, so assume $\gamma > 0$. Choose unit vectors $x, y \in \mathbb{C}^n \otimes \mathbb{C}^m$ such that

$$F[A]x \;=\; \gamma y, \qquad F[A]^* y \;=\; \gamma x. \tag{5.1}$$

Such a pair exists by the finite-dimensional singular value decomposition.

Identify $\mathbb{C}^n \otimes \mathbb{C}^m \cong (\mathbb{C}^n)^m = \mathbb{C}^{nm}$ via the standard basis $e_1, \dots, e_m$ of $\mathbb{C}^m$. Any vector $x \in \mathbb{C}^n \otimes \mathbb{C}^m$ can be written uniquely as $x = \sum_{i=1}^m Xe_i \otimes e_i$ for an $n \times m$ matrix X. Similarly, write $y = \sum_{i=1}^m Ye_i \otimes e_i$ with Y an $n \times m$ matrix. The unit-length conditions become

$$\|X\|_2^2 = \|x\|^2 = 1, \qquad \|Y\|_2^2 = \|y\|^2 = 1$$

(here and below $\|X\|_2^2 = \operatorname{tr}(X^*X)$).

Define the $m \times m$ matrix functions

$$P = X^*RX, \quad Q = Y^*RY, \quad M = Y^*RX, \quad L = X^*RY. \tag{5.2}$$

Their Laurent coefficients in the variable λ^{-1} will be denoted with subscripts: $M_k = Y^*b_k(A)X$, $P_k = X^*b_k(A)X$, $Q_k = Y^*b_k(A)Y$, $L_k = X^*b_k(A)Y$. Since $\|b_k(A)\| \le C\theta^k$ (Theorem 4.1), all four series converge absolutely and uniformly on $\mathbb{T}$, and all four are square-summable in the Hilbert-Schmidt norm.

Important bookkeeping at $k = 0$. Since $b_0(A) = I$,

$$P_0 = X^*X, \quad Q_0 = Y^*Y, \quad L_0^* = M_0. \tag{5.3}$$

In particular P_0 and Q_0 are Hermitian and positive semidefinite.

Finally, set

$$a^2 \;=\; \sum_{k\ge 0} \|M_k\|_2^2, \qquad \text{finite and } a > 0. \tag{5.4}$$

The finiteness is from the geometric decay of $b_k(A)$. The positivity $a > 0$ follows from applying the cross-functional bound below to $G = F$: the value $y^*F[A]x = \gamma > 0$ cannot be achieved by a Cauchy-Schwarz bound with a zero right-hand side.

Step A. The off-diagonal functional

For any $M_m(\mathbb{C})$-valued polynomial $G(z) = \sum_{k=0}^N b_k(z)C_k$ written in the Faber basis, direct expansion in the tensor product gives

$$y^*G[A]x \;=\; \sum_{k=0}^{N} \sum_{i,j=1}^{m} (M_k)_{ij}\,(C_k)_{ij}. \tag{5.5}$$

Let us verify (5.5) once (it is used three more times). Since $x = \sum_j Xe_j \otimes e_j$ and $G[A] = \sum_k b_k(A) \otimes C_k$,

$$G[A]x \;=\; \sum_{k,j} (b_k(A)Xe_j) \otimes (C_k e_j).$$

Pairing with $y = \sum_i Ye_i \otimes e_i$:

$$y^*G[A]x \;=\; \sum_{k,i,j} (Ye_i)^* b_k(A)(Xe_j) \cdot e_i^* C_k e_j \;=\; \sum_{k,i,j} (M_k)_{ij}\,(C_k)_{ij},$$

as claimed. Equation (5.5) is a scalar bilinear pairing of the coefficient arrays $(M_k)_{ij}$ and $(C_k)_{ij}$.

Cauchy-Schwarz on (5.5) gives

$$|y^*G[A]x| \;\le\; \left(\sum_{k\ge 0}\|M_k\|_2^2\right)^{1/2}\left(\sum_{k=0}^{N}\|C_k\|_2^2\right)^{1/2} \;=\; a\left(\sum_{k=0}^{N}\|C_k\|_2^2\right)^{1/2}. \tag{5.6}$$

Plugging in $G = F$ and using $y^*F[A]x = \gamma$ from (5.1) shows that $a > 0$.

Step B. Two ordered tests

Polynomial evaluation with matrix coefficients respects multiplication in the given order. If $F(z) = \sum_i D_i z^i$ and $G(z) = \sum_j E_j z^j$, then

$$(FG)[A] \;=\; \sum_{i,j} A^{i+j}\otimes D_iE_j \;=\; F[A]G[A].$$

Similarly $(GF)[A] = G[A]F[A]$ when factors are in the reversed order. Multiplying (5.1) on the left by suitable pieces,

$$y^*(FG)[A]x \;=\; y^*F[A]G[A]x \;=\; \gamma\, x^*G[A]x, \qquad y^*(GF)[A]x \;=\; \gamma\, y^*G[A]y. \tag{5.7}$$

So the diagonal functionals $x^*G[A]x$ and $y^*G[A]y$ can be turned into the off-diagonal functional by ordering the two factors differently.

Write the two products in the Faber basis:

$$FG \;=\; \sum_{k\ge 0} b_kU_k, \qquad GF \;=\; \sum_{k\ge 0} b_kV_k,$$

with finitely many nonzero coefficient matrices $U_k, V_k \in M_m(\mathbb{C})$. Define the boundary norm

$$\|H\|_\partial^2 \;=\; \int_{\mathbb{T}} \|H(h(\mu))\|_2^2\, d\tau(\mu), \qquad H \in M_m(\mathbb{C})[z].$$

Because $\|F(z)\| \le 1$ on the boundary, submultiplicativity of the pointwise operator norm gives

$$\|FG\|_\partial \;\le\; \|F\|_\partial\,\|G\|_\partial \;\le\; \|G\|_\partial, \qquad \|GF\|_\partial \;\le\; \|G\|_\partial.$$

(The two inequalities $\|F(z)G(z)\|_2 \le \|F(z)\|\|G(z)\|_2$ and $\|G(z)F(z)\|_2 \le \|G(z)\|_2\|F(z)\|$ are used in their respective orders.)

Apply (5.6) to FG and GF, and use (5.7), the lower coefficient estimate from Proposition 3.2, and the boundary multiplier bounds:

$$\gamma\,|x^*G[A]x| = |y^*(FG)[A]x| \le a\left(\sum_k \|U_k\|_2^2\right)^{1/2} \le a\|FG\|_\partial \le a\|G\|_\partial,$$

$$\gamma\,|y^*G[A]y| = |y^*(GF)[A]x| \le a\left(\sum_k \|V_k\|_2^2\right)^{1/2} \le a\|GF\|_\partial \le a\|G\|_\partial.$$

The upper coefficient estimate for G in Proposition 3.2 turns $\|G\|_\partial$ into a norm of the coefficient array: $\|G\|_\partial \le (\|C_0\|_2^2 + 2\sum_{k\ge 1}\|C_k\|_2^2)^{1/2}$. Therefore

$$\max\{|x^*G[A]x|, |y^*G[A]y|\} \;\le\; \frac{a}{\gamma}\left(\|C_0\|_2^2 \;+\; 2\sum_{k\ge 1}\|C_k\|_2^2\right)^{1/2}. \tag{5.8}$$

This is the crucial estimate that produces the weight of two.

Step C. Weighted dual norms

The functional $x^*G[A]x$ pairs C_k against P_k (repeat the calculation of (5.5), but with y replaced by x; the coefficient array is $P_k = X^*b_k(A)X$). Similarly $y^*G[A]y$ pairs C_k against Q_k. So (5.8) constrains the size of the sequences (P_k) and (Q_k), with weights that reverse the ones in (5.8).

Concretely, for each $N \geq 0$ set

$$d_N^{(P)} \;=\; \|P_0\|_2^2 \;+\; \tfrac{1}{2}\sum_{k=1}^{N} \|P_k\|_2^2.$$

Test (5.8) with the coefficient sequence

$$C_0 = \overline{P_0}, \qquad C_k = \overline{P_k}/2 \;(1 \leq k \leq N), \qquad C_k = 0 \;(k > N),$$

where the bar denotes entrywise complex conjugation. The scalar paired value is exactly $d_N^{(P)}$. The weighted norm on the right is $\|C_0\|_2^2 + 2\sum_{k=1}^{N} \|C_k\|_2^2 = \|P_0\|_2^2 + \tfrac{1}{2}\sum_{k=1}^{N} \|P_k\|_2^2 = d_N^{(P)}$. So

$$d_N^{(P)} \;\leq\; \frac{a}{\gamma}\,(d_N^{(P)})^{1/2}.$$

If $d_N^{(P)} = 0$ this is trivial. Otherwise divide and square: $d_N^{(P)} \leq a^2/\gamma^2$. Letting $N \to \infty$:

$$\|P_0\|_2^2 + \tfrac{1}{2}\sum_{k\geq 1} \|P_k\|_2^2 \;\leq\; \frac{a^2}{\gamma^2}, \qquad \|Q_0\|_2^2 + \tfrac{1}{2}\sum_{k\geq 1} \|Q_k\|_2^2 \;\leq\; \frac{a^2}{\gamma^2}. \tag{5.9}$$

The same finite test with Q_k in place of P_k gives the second bound.

Step D. Boundary L^2 norms of the blocks

Now compute the square-integrated Hilbert-Schmidt norms of the three extreme matrix fields $P + P^*$, $Q + Q^*$, and $M + L^*$ on the circle.

Diagonal blocks. The function $P + P^*$ has Laurent expansion

$$(P + P^*)(\lambda) \;=\; 2P_0 \;+\; \sum_{k\geq 1} P_k\,\lambda^{-k} \;+\; \sum_{k\geq 1} P_k^*\,\lambda^{k}$$

where we used $P_0^* = P_0$ and $\overline{\lambda}^k = \lambda^{-k}$ on $|\lambda| = 1$. The three groups (constant, negative powers, positive powers) have disjoint Fourier support, so Parseval gives

$$\int_{\mathbb{T}} \|P + P^*\|_2^2\, d\tau(\lambda) \;=\; \|2P_0\|_2^2 \;+\; \sum_{k\geq 1} \|P_k\|_2^2 \;+\; \sum_{k\geq 1} \|P_k^*\|_2^2 \;=\; 4\|P_0\|_2^2 \;+\; 2\sum_{k\geq 1} \|P_k\|_2^2 \;=\; 4d_\infty^{(P)},$$

where $d_\infty^{(P)} = \|P_0\|_2^2 + \tfrac{1}{2}\sum_{k\geq 1} \|P_k\|_2^2$ is the limit of the sequence $d_N^{(P)}$ from Step C. Combined with (5.9),

$$\int_{\mathbb{T}} \|P + P^*\|_2^2\, d\tau \;\leq\; 4a^2/\gamma^2, \qquad \int_{\mathbb{T}} \|Q + Q^*\|_2^2\, d\tau \;\leq\; 4a^2/\gamma^2. \tag{5.10}$$

The second inequality follows from the second half of (5.9) by the same calculation with Q in place of P.

Cross block. The function $M + L^*$ has Laurent expansion

$$(M + L^*)(\lambda) \;=\; \sum_{k\geq 0} M_k\, \lambda^{-k} \;+\; \sum_{k\geq 0} L_k^*\, \lambda^k.$$

The constant coefficient is $M_0 + L_0^* = 2M_0$, using $L_0^* = M_0$ from (5.3). Grouping by Fourier mode,

$$\int_{\mathbb{T}} \|M + L^*\|_2^2\, d\tau \;=\; \|2M_0\|_2^2 \;+\; \sum_{k\geq 1} \|M_k\|_2^2 \;+\; \sum_{k\geq 1} \|L_k^*\|_2^2.$$

Since $\|L_k^*\|_2 = \|L_k\|_2$ (the Hilbert-Schmidt norm is invariant under adjoint) and every term is nonnegative,

$$\int_{\mathbb{T}} \|M + L^*\|_2^2\, d\tau \;\geq\; 4\|M_0\|_2^2 \;+\; \sum_{k\geq 1} \|M_k\|_2^2 \;\geq\; \sum_{k\geq 0} \|M_k\|_2^2 \;=\; a^2. \tag{5.11}$$

The final inequality holds because the coefficient of $4\|M_0\|_2^2$ exceeds the coefficient of $\|M_0\|_2^2$; dropping the $\|L_k\|_2^2$ terms only makes the left smaller.

Step E. Positive block comparison

The proof now closes with linear algebra.

From Theorem 4.1, $R(\lambda) + R(\lambda)^* \succeq 0$ for every $\lambda \in \mathbb{T}$. Compress by the $n \times 2m$ matrix $[X\ Y]$. The $2m \times 2m$ matrix of blocks

$$\Delta(\lambda) \;=\; \begin{pmatrix} P + P^* & L + M^* \\ M + L^* & Q + Q^* \end{pmatrix}(\lambda) \;=\; \begin{pmatrix} X \\ Y \end{pmatrix}^* (R + R^*)(\lambda) \begin{pmatrix} X & Y \end{pmatrix} \;\succeq\; 0$$

is positive semidefinite for every λ. A compression of a positive semidefinite matrix is positive semidefinite, so this is immediate.

Let $J = \operatorname{diag}(I_m, -I_m)$. If $\Delta \succeq 0$, then $J\Delta J \succeq 0$ too (the conjugation by J keeps positivity). If two matrices S and T are positive semidefinite, then $\operatorname{tr}(ST) \geq 0$: writing $S = AA^*$ and $T = BB^*$, $\operatorname{tr}(ST) = \operatorname{tr}(AA^*BB^*) = \operatorname{tr}((A^*B)^*(A^*B)) = \|A^*B\|_2^2 \geq 0$. So

$$0 \;\leq\; \operatorname{tr}(\Delta(J\Delta J)) \;=\; \operatorname{tr}((\Delta J)^2).$$

Compute $(\Delta J)^2$ directly. $\Delta J = \begin{pmatrix} P + P^* & -(L + M^*) \\ M + L^* & -(Q + Q^*) \end{pmatrix}$. Then

$$\operatorname{tr}\big((\Delta J)^2\big) \;=\; \operatorname{tr}\big((P+P^*)^2\big) \;-\; \operatorname{tr}\big((L+M^*)(M+L^*)\big) \;-\; \operatorname{tr}\big((M+L^*)(L+M^*)\big) \;+\; \operatorname{tr}\big((Q+Q^*)^2\big).$$

Since $\operatorname{tr}((AB)^*) = \overline{\operatorname{tr}(AB)}$ but also $(L + M^*)(M + L^*)$ and its adjoint $(M + L^*)(L + M^*)$ have the same trace (they are adjoints, and their traces are complex conjugates; both equal $\operatorname{tr}(AA^*)$ with $A = L + M^*$, which is real nonnegative), the two middle terms sum to

$$-\operatorname{tr}\big((L + M^*)(M + L^*) + (M + L^*)(L + M^*)\big) \;=\; -2\operatorname{tr}(L + M^*)^*(L + M^*) \;=\; -2\|M + L^*\|_2^2.$$

Putting it all together:

$$0 \;\leq\; \|P + P^*\|_2^2 \;+\; \|Q + Q^*\|_2^2 \;-\; 2\|M + L^*\|_2^2. \tag{5.12}$$

This holds for every $\lambda \in \mathbb{T}$.

Integrate (5.12) over $\mu \in \mathbb{T}$, use (5.10) for the two diagonal terms and (5.11) for the cross term:

$$0 \;\le\; 4a^2/\gamma^2 + 4a^2/\gamma^2 \;-\; 2a^2 \;=\; 8a^2/\gamma^2 \;-\; 2a^2.$$

So

$$2a^2 \;\le\; 8a^2/\gamma^2.$$

Since $a > 0$, divide by a^2: $\gamma^2 \le 4$, hence $\gamma \le 2$.

This finishes the proof that $\|F[A]\| \le 2$. □

Remarks on the argument

1. *Where the* 2 *comes from.* The factor of two in the upper coefficient estimate (3.4) is multiplied through Step E until it becomes $\gamma^2 = 4$. Removing the factor of two would give $\gamma^2 \le 2$, and that is false in general (the sharpness example in Section 6 has $\gamma = 2$).

2. *Where convexity enters.* The only place convexity is used is in the support inequality (2.3), which in turn gives the positivity of K (Theorem 2.1) and the positivity of $R + R^*$ (Theorem 4.1). If Ω were not convex, both positivity statements could fail and the argument would not go through.

3. *Why scalar proofs do not suffice.* In a scalar argument ($m = 1$), one would use $FG = GF$ freely. Here, the two products FG and GF are genuinely different $M_m(\mathbb{C})$-valued polynomials. The whole point of Step B is to use these two different products separately, obtaining bounds on $x^*G[A]x$ and $y^*G[A]y$ that share the same cross-functional a.

6 Enclosing domains and the general case

We have proved $\|F[A]\| \le 2$ under the hypothesis $W(A) \subset \Omega$ with Ω an analytic convex domain. We now remove that hypothesis: for an arbitrary compact convex $W(A)$ we construct analytic convex neighborhoods Ω_j shrinking to $W(A)$, apply the previous section, and pass to the limit.

Analytic convex outer approximation

Lemma 6.1 (Analytic convex outer approximation)**.** *Let $K \subset \mathbb{C}$ be nonempty, compact, and convex. There exist bounded open convex domains $\Omega_j \subset \mathbb{C}$ with regular real-analytic Jordan boundaries such that $K \subset \Omega_j$ for every j and*

$$\sup_{z \in \overline{\Omega}_j} \operatorname{dist}(z, K) \;\xrightarrow[j\to\infty]{}\; 0.$$

Proof. The construction is a smooth convex interpolation between K and a small thickening. Fix $0 < \delta < 1$. Choose a finite δ-net D_δ of unit directions that includes $1, -1, i, -i$. Let $s_K(\nu) = \max_{z \in K} \Re(\overline{\nu} z)$ be the support function of K. Choose $t_\delta > 0$ small enough that $|D_\delta| e^{-t_\delta \delta} < 1$.

Define the smooth, strictly convex function

$$\Phi_\delta(z) \;=\; \sum_{\nu \in D_\delta} \exp\bigl(t_\delta \,(\Re(\overline{\nu} z) - s_K(\nu) - \delta)\bigr)$$

and set $\Omega_\delta = \{z \in \mathbb{C} : \Phi_\delta(z) < 1\}$.

(i) $K \subset \Omega_\delta$. For $z \in K$ and any ν, $\Re(\overline{\nu} z) \le s_K(\nu)$, so each exponential is at most $e^{-t_\delta \delta}$, and the sum is at most $|D_\delta| e^{-t_\delta \delta} < 1$.

(ii) $\overline{\Omega}_\delta$ lies in a fixed rectangle. If $\Phi_\delta(z) \le 1$, then every summand is ≤ 1 (since each is nonnegative), hence $\Re(\overline{\nu} z) \le s_K(\nu) + \delta$ for every $\nu \in D_\delta$. In particular, for $\nu = 1$ and $\nu = -1$, we get $\Re z$ in $[\min_{w \in K} \Re w - \delta, \max_{w \in K} \Re w + \delta]$; and similarly the imaginary part. So all $\overline{\Omega}_\delta$ for $0 < \delta < 1$ sit inside one fixed bounded rectangle $R \subset \mathbb{C}$.

(iii) Φ_δ is strictly convex with positive definite Hessian. The Hessian is a sum of positive multiples of $\nu\nu^{\mathsf{T}}$ (outer products), where ν ranges over D_δ. Each $\nu\nu^{\mathsf{T}}$ is positive semidefinite; the four directions $1, -1, i, -i$ together span $\mathbb{R}^2$, so the sum is strictly positive definite. Hence Φ_δ is strictly convex and real analytic.

(iv) Level set is regular and a Jordan curve. If z_0 were a critical point of Φ_δ with $\Phi_\delta(z_0) = 1$, then by convexity Φ_δ would attain a global minimum at z_0, contradicting $\Phi_\delta(\text{any } z \in K) < 1$. So $\nabla \Phi_\delta \neq 0$ on $\Phi_\delta^{-1}(1)$. The implicit function theorem makes that level a regular real-analytic curve. Pick an interior point $w_0 \in \Omega_\delta$ (any point of K will do). Each ray from w_0 crosses $\partial\Omega_\delta$ exactly once, because Φ_δ is strictly convex and bounded. Radial projection gives a continuous bijection $\partial\Omega_\delta \to \mathbf{S}^1$, so $\partial\Omega_\delta$ is a Jordan curve and $\overline{\Omega}_\delta = \{\Phi_\delta \le 1\}$.

(v) Distance to K. For $z \in \overline{\Omega}_\delta$, let $w \in K$ be the unique nearest point of K and put $d = |z - w|$. If $z \in K$, then $d = 0$. Otherwise $d > 0$; set $\nu = (z - w)/d$, a unit vector. Choose $\mu \in D_\delta$ with $|\mu - \nu| \le \delta$, and choose $v \in K$ attaining $s_K(\mu) = \Re(\overline{\mu} v)$.

Using minimality of w, for every $w' \in K$ we have $\Re(\overline{\nu}(w' - w)) \le 0$ (the real part of the derivative of $|z - w|^2$ along $w + t(w' - w)$ at $t = 0$). In particular $\Re(\overline{\nu}(v - w)) \le 0$ and $\Re(\overline{\mu}(w - v)) \le 0$.

Now compute:

- Since $|\mu - \nu| \le \delta$, $\Re(\overline{\mu}\nu) \ge 1 - \delta$ and hence $\Re(\overline{\mu}(z - w)) = d\Re(\overline{\mu}\nu) \ge (1 - \delta)\, d$.
- Since $\Re(\overline{\mu}(v - w)) \ge 0$ (support inequality, since v maximizes $\Re(\overline{\mu} \cdot)$ on K) and since $\Phi_\delta(z) \le 1$ forces $\Re(\overline{\mu} z) \le \Re(\overline{\mu} v) + \delta$, we get $\Re(\overline{\mu}(z - v)) \le \delta$.
- From $\Re(\overline{\nu}(v - w)) \le 0$, together with $|\mu - \nu| \le \delta$, we get $\Re(\overline{\mu}(v - w)) \le \delta |v - w|$.

Hence

$$(1 - \delta)\, d \;\le\; \Re(\overline{\mu}(z - w)) \;=\; \Re(\overline{\mu}(z - v)) + \Re(\overline{\mu}(v - w)) \;\le\; \delta + \delta\, |v - w| \;\le\; \delta(1 + \operatorname{diam} K).$$

Points of K have distance zero, so

$$\sup_{z \in \overline{\Omega}_\delta} \operatorname{dist}(z, K) \;\le\; \frac{\delta(1 + \operatorname{diam} K)}{1 - \delta}.$$

Taking $\delta = \delta_j = 1/(j + 1)$ and $\Omega_j = \Omega_{\delta_j}$ completes the construction. □

Proof of the main theorem

Proof of Theorem 1.1. First recall the Toeplitz–Hausdorff convexity theorem [17, 11]: $W(A)$ is convex.

Compactly. $W(A) = \{u^* A u : u^* u = 1\}$ is the image of the unit sphere (compact) under a continuous map, hence is compact.

Convexly. Take any $u, v \in \mathbb{C}^n$ with $|u| = |v| = 1$. We want to show that the segment $[u^* A u, v^* A v]$ lies in $W(A)$. Let $V = \operatorname{span}\{u, v\}$, of complex dimension 1 or 2.

If $\dim V = 1$, then $v = e^{i\theta} u$ and $v^* A v = u^* A u$, so the segment is trivial.

If $\dim V = 2$, work in an orthonormal basis $\{e_1, e_2\}$ of V, and let B be the 2×2 matrix of $P_V A|_V$ in this basis. For a unit vector $x = x_1 e_1 + x_2 e_2$, writing $X = 2\Re(\overline{x_1} x_2)$, $Y = 2\Im(\overline{x_1} x_2)$, $Z = |x_1|^2 - |x_2|^2$, one has $X^2 + Y^2 + Z^2 = 1$ (because $4|x_1 x_2|^2 + (|x_1|^2 - |x_2|^2)^2 = (|x_1|^2 + |x_2|^2)^2 = 1$), so (X, Y, Z) ranges over the unit sphere $\mathbf{S}^2 \subset \mathbb{R}^3$, and $x^* B x$ is a real-linear function $f(X, Y, Z)$ into $\mathbf{C} \cong \mathbb{R}^2$. The linear part of f has a one-dimensional kernel. Every point of the unit ball can be pushed along the kernel direction onto the sphere without changing f, so $f(\mathbf{S}^2) = f(\overline{\mathbb{B}^3})$. The image of a ball under a linear map is convex, so $f(\mathbf{S}^2)$ is convex. In particular, the segment $[u^* A u, v^* A v] \subset f(\mathbf{S}^2) \subset W(A)$.

So $W(A)$ is compact and convex.

Enclosing. Apply Lemma 6.1 with $K = W(A)$. Get analytic convex domains Ω_j with $W(A) \subset \Omega_j$ and $\sup_{\overline{\Omega}_j} \operatorname{dist}(\cdot, W(A)) \to 0$. In particular every $\overline{\Omega}_j$ lies inside the fixed bounded rectangle from Lemma 6.1(*ii*).

Uniform bound. Fix one Ω_j. Let $M_j = \sup_{z \in \overline{\Omega}_j} \|P(z)\|$. Two cases.

If $M_j = 0$, then P vanishes on Ω_j (open), so P is the zero polynomial and $P[A] = 0$.

If $M_j > 0$, multiply P by $1/M_j$. The normalized polynomial satisfies $\|P/M_j\| \le 1$ on $\partial\Omega_j$, so Theorem 5.1 applies and gives $\|(P/M_j)[A]\| \le 2$, i.e., $\|P[A]\| \le 2M_j$.

In either case, $\|P[A]\| \le 2M_j$.

Shrinking. Since $\overline{\Omega}_j$ converges to $W(A)$ in Hausdorff distance and all $\overline{\Omega}_j$ lie in a fixed compact set, uniform continuity of the continuous function $z \mapsto \|P(z)\|$ implies $M_j \to \sup_{z \in W(A)} \|P(z)\| = \max_{W(A)} \|P\|$. Passing to the limit in $\|P[A]\| \le 2M_j$,

$$\|P[A]\| \;\le\; 2 \max_{z \in W(A)} \|P(z)\|.$$

This is exactly (1.3). □

Sharpness of the constant 2

Sharpness in Theorem 1.1. Take $n = 2$, $m = 1$, $d = 1$, $P(z) = z$, and $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$. We already saw in the Introduction that $W(A)$ is the closed unit disk, and that $\|A\| = 2$; so $\|P[A]\| = 2$ and $\max_{W(A)} |P(z)| = 1$. The ratio is

$$\frac{\|P[A]\|}{\max_{W(A)} \|P(z)\|} \;=\; \frac{2}{1} \;=\; 2.$$

Hence no smaller constant works uniformly in n, m, d. □

The constant 2 is achieved by a very asymmetric matrix: A is 2-step nilpotent, $W(A)$ is a disk (not the spectrum), and the ratio $\|A\| / \max_{W(A)} |z|$ is exactly 2. This is why the numerical range, not the spectrum, is the right set: the spectrum of A is $\{0\}$, and no bound $\|P[A]\| \le C \max_{\sigma(A)} |P|$ would hold for any finite C.

A The exterior conformal collar

This appendix supplies the classical conformal input used in Section 2. Everything here is covered by Ahlfors [2]; we spell out the two ingredients that are rarely stated explicitly: real-analytic boundary continuation and global injectivity on a small exterior collar.

Lemma A.1 (Exterior coordinate and a univalent collar)**.** *Let $\Omega \subset \mathbb{C}$ be a bounded convex domain with regular real-analytic Jordan boundary. Then there are $0 < r_0 < 1$ and a map h which is meromorphic and one-to-one on $\{\lambda : |\lambda| > r_0\} \cup \{\infty\}$, has a simple pole at ∞, and maps $|\lambda| > 1$ conformally onto $\mathbb{C} \setminus \overline{\Omega}$, and $|\lambda| = 1$ onto $\partial\Omega$. The expansion at infinity is*

$$h(\lambda) \;=\; c\lambda + c_0 + \sum_{j\geq 1} c_j \lambda^{-j}, \qquad c \neq 0.$$

For λ on the unit circle, $q(\lambda) = \lambda h'(\lambda)$ is a nonzero outward normal vector at $h(\lambda)$, and the support inequality (2.3) holds.

Proof. Step 1: A bounded simply connected model. Choose a point $z_* \in \Omega$ and apply $z \mapsto 1/(z - z_*)$ (with ∞ mapped to 0). The spherical exterior $(\mathbb{C}\setminus\overline{\Omega})\cup\{\infty\}$ maps onto a bounded simply connected domain U containing 0. Because Ω is convex, each ray from z_* meets $\partial\Omega$ in a unique point; after inversion, the exterior portions of those rays become a domain that is star-shaped about 0. Its boundary is a regular real-analytic Jordan curve: the inversion of a regular real-analytic curve is again one.

Step 2: Riemann mapping. By the Riemann mapping theorem [2, Ch. 6, §1.1, Thm. 1], there is a conformal bijection $g : \mathbb{D} \to U$ with $g(0) = 0$. By Carathéodory's Jordan-domain theorem [2, Ch. 6, §1.2], g extends to a homeomorphism $\overline{\mathbb{D}} \to \overline{U}$.

Step 3: Local real-analytic continuation across the boundary. A regular real-analytic boundary arc has a real-analytic parametrisation with nonzero derivative. Its holomorphic continuation and the holomorphic inverse function theorem give local coordinates in which the arc is a real interval and the domain is one side of that interval. Fix such coordinates at a boundary point $\zeta \in \partial\mathbb{D}$ and its image $p = g(\zeta) \in \partial U$. Continuity of the boundary extension allows shrinking the source neighborhood so that g maps its intersection with $\overline{\mathbb{D}}$ into the target chart. In these coordinates, g is holomorphic on a half-disk, continuous on the real interval, and real-valued there. The Schwarz reflection principle [2, Ch. 4, §6.5; Ch. 6, §§1.3–1.4] extends g holomorphically across the interval. Apply the same procedure to g^{-1}. Shrinking neighborhoods allows composing the reflected maps. Their compositions equal the identity on the original domain side, and hence throughout a smaller neighborhood by the identity theorem. So the continued g has nonzero derivative at every boundary point.

Step 4: Patching local continuations. For each $\zeta \in \partial\mathbb{D}$, choose a disk $B(\zeta, R_\zeta)$ on which such a continuation $\tilde{g}_\zeta$ exists, agreeing with g on $B(\zeta, R_\zeta) \cap \mathbb{D}$. Cover the compact circle by finitely many smaller disks $V_i = B(\zeta_i, R_i/3)$; write $U_i = B(\zeta_i, R_i)$. If $V_i \cap V_j \neq \varnothing$ and $R_i \geq R_j$, then $|\zeta_i - \zeta_j| < (R_i + R_j)/3 \leq 2R_i/3 < R_i$, so the two larger disks overlap in a connected lens $U_i \cap U_j$ that contains ζ_j and meets $\mathbb{D}$. On $\mathbb{D} \cap U_i \cap U_j$, both $\tilde{g}_i$ and $\tilde{g}_j$ agree with g, hence with each other. By the identity theorem they agree on the whole lens $U_i \cap U_j$. So the local extensions patch with g to a single holomorphic function on $\mathbb{D} \cup \bigcup_i V_i$, which is a neighborhood of the closed disk.

Step 5: Global injectivity on a small neighborhood. The extension is injective on a (possibly smaller) neighborhood of the closed disk. Otherwise there would be sequences $(w_n), (w_n')$ with $w_n \neq w_n'$, $w_n, w_n' \to$ (a neighborhood of closed disk), and $\tilde{g}(w_n) = \tilde{g}(w_n')$. Passing to subsequences inside the compact closed disk, both go to a common limit w^* in the closed disk. Since $\tilde{g}$ is continuous and injective on the closed disk (g is a homeomorphism on $\overline{\mathbb{D}}$), and $\tilde{g}(w_n) = \tilde{g}(w_n')$, we must eventually have w_n, w_n' in a common small disk about w^* on which $\tilde{g}$ is injective, contradiction. By compactness, the (injective) neighborhood is open and contains a disk $\{w : |w| < R_0\}$ for some $R_0 > 1$.

Step 6: Assembly. The function g is univalent on $\{w : |w| < R_0\} \supset \overline{\mathbb{D}}$. Its only zero there is 0,

and that zero is simple ($g'(0) \neq 0$ by conformality). Set

$$h(\lambda) \;=\; z_* + \frac{1}{g(1/\lambda)}, \qquad r_0 \;=\; R_0^{-1}.$$

Then h is meromorphic on $\{|\lambda| > r_0\} \cup \{\infty\}$, has a simple pole at ∞ (from the simple zero of g at 0), maps $|\lambda| > 1$ conformally onto $\mathbb{C} \setminus \overline{\Omega}$ (from g mapping $\mathbb{D}$ to U and $z \mapsto z_* + 1/z$ undoing the inversion), and maps $|\lambda| = 1$ onto $\partial\Omega$. The expansion $h(\lambda) = c\lambda + c_0 + O(1/\lambda)$ has $c = 1/g'(0) \neq 0$.

Step 7: The support inequality. At $\lambda = e^{i\theta} \in \mathbb{T}$, the radial and angular derivatives of h are $q(\lambda) = \lambda h'(\lambda)$ and $iq(\lambda)$, respectively. They are perpendicular and nonzero. Increasing the radial parameter $|\lambda|$ traces out the exterior of $\overline{\Omega}$, so $q(\lambda)$ points outward. Since Ω is convex with a smooth boundary, the supporting-line property holds at $h(\lambda)$: for every $z \in \overline{\Omega}$,

$$\Re\Big(\overline{q(\lambda)}\big(h(\lambda) - z\big)\Big) \;\geq\; 0.$$

This is exactly (2.3). □

Sanity check for the disk

For $\Omega = \mathbb{D}$ (the unit disk), the exterior coordinate is $h(\lambda) = \lambda$. Then $q(\lambda) = \lambda$, $b_k(z) = z^k$, and $s(\lambda, \mu) = 0$. The kernel K is 1, and the marginal identities are trivial. The support inequality reads $\Re(\overline{\lambda}(\lambda - z)) = 1 - \Re(\overline{\lambda} z) \geq 0$ for $|z| \leq 1$, $|\lambda| = 1$, which is just Cauchy–Schwarz.

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